A simple model of lateral congestion in fairways

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1 Introduction

This document describes a model of vessel traffic in a fairway. The model considers unidirectional traffic and can be applied to bidirectional flows by modeling each direction separately. Differently from multi-lane road traffic, there may not be lateral space constraints (limitation on the number of lanes) – vessels may spread out laterally. The model computes the required fairway geometry occupied by the vessels. A major current simplification is that vessel speeds are assumed to be constant, such that lateral evasive/overtaking maneuvers are the only way to resolve conflicts between vessels moving in the same direction at different speeds.

2 Method

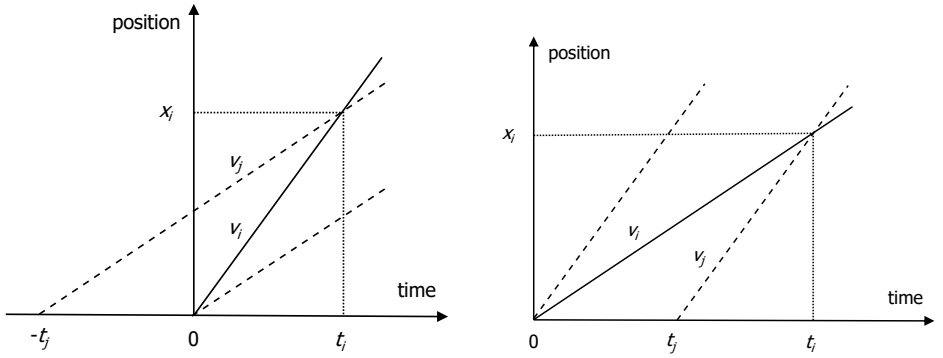
Vessel classes

Consider a vessel fleet that is grouped into classes $i = 1, \ldots, J$. The speed, length, and draught of class i is v_i, d_i, and g_i, respectively. The geometric parameters include possible safety margins. Consider a single direction of a bidirectional fairway. Vessels of class i enter the fairway at rate λ_i (in vessels per time unit). Divide the fairway into hypothetical lanes $l = 1, 2, ...$, where 1 is the innermost (and deepest) lane. We assume that vessels travel at fixed speeds and resolve conflicts by lateral maneuvers. We are interested in the resulting stationary lateral pattern of vessels across lanes.

Model inputs

Vessels are grouped into classes because the model describes traffic in terms of class-specific arrival rates. While an individual observed vessel occurs only once, grouping vessels with similar characteristics allows an arrival rate λ_i to be estimated for each class.

For each vessel class i, the model requires the arrival rate λ_i [vessels per unit of time], speed v_i [m per unit of time], length d_i [m], width w_i [m], and draught g_i [m]. Length and width may include safety margins. The classes are prioritized by draught, with the largest draught having the highest priority. The



(a) Class i overtakes class j (b) Class i yields to class j

Figure 1: Space/time diagrams for overtaking and yielding

calculation also requires a selected percentile of the vessel-width distribution for determining the width of each used lane.

Overtaking and yielding

Consider vessel class i on lane l. It wants to move through lane l at rate $\lambda'_{i\to l}$ (which may be less than λ_i for reasons explained further below). Assume that there are already vessels from some other class $j \neq i$ on that lane, and assume that class j has priority over class i. A vessel class j has priority over class i if a conflict between class i and j has to be resolved by the class-i vessel. We index classes according to their (unique) priority, with class 1 having the highest and class J the lowest priority.

We would like to model how a class-i vessel maneuvers through that lane, given three options: (i) advance on the lane, given that no conflict arises, (ii) overtake a slower class-j vessel on the right, (iii) yield for a faster class-j vessel to the right. These rules are consistent with the assumption that vessels stay as far left (i.e. centered) as possible unless forced to do otherwise.

Consider a single class-i vessel that moves at speed v_i along lane l, advancing from position zero to x_i within t_i time units. Assume that the lane is also used by class-j vessels, all which have priority over class i.

Overtaking. This case is illustrated in Figure 1(a). All slower class-j vessels (moving at speed $v_j < v_i$) being overtaken by the class-i vessel between times zero and t_i time units have crossed the zero position during the time interval $[-t_j, 0]$. Geometrical considerations yield $t_j = x_i \cdot \left(\frac{1}{v_j} - \frac{1}{v_i}\right)$. Letting $\lambda_{j\to l}$ be the rate at which class-j vessels pass through lane l, the class-i vessel overtakes $\lambda_{j\to l} t_j$ class-j vessels during t_i time units. A single overtaking maneuver is assumed to take a fixed (vessel-class specific) time, set here to $\theta_{ij} = \frac{d_i + d_j}{v_i - v_j}$. This is the time it takes for a class-i vessel that

starts behind a class-j vessels to move past that vessel on lane $l+1$ and to push back into lane l safely in front of the class-j vessel. Out of the t_i time units it takes the class-i vessel to advance by x_i space units, it spends $\lambda_{j\to l} t_j \cdot \theta_{ij}$ time units overtaking class-j vessels. The resulting *share* of time spent by the class-i vessel overtaking class-j vessels that move on l is

$$\sigma_{ij|l} \quad = \quad \frac{\lambda_{j\to l}(d_i + d_j)}{v_j}. \tag{1}$$

Yielding. This case is illustrated in Figure 1(b). All faster class-j vessels (moving at speed $v_j > v_i$) catching up on the class-i vessel during the time interval $[0, t_i]$ cross the zero position during the time interval $[0, t_j]$. Geometrical considerations yield $t_j = x_i \cdot \left(\frac{1}{v_i} - \frac{1}{v_j}\right)$, meaning that the class-i vessel has to yield to $\lambda_{j\to l} t_j$ class-j vessels, by temporarily changing to lane $l+1$. A single yielding maneuver is assumed to equal the time it takes a class-j vessel to overtake a class-i vessel: $\theta_{ji} = \frac{d_i + d_j}{v_j - v_i}$. Out of the t_i time units it takes the class-i vessel to advance by x_i space unit, it spends $\lambda_{j\to l} t_j \theta_{ji}$ time units yielding to class-j vessels. The resulting *share* of time spent by class-i yielding to class-j vessels that move on l is

$$\sigma_{ij|l} \quad = \quad \frac{\lambda_{j\to l}(d_i + d_j)}{v_j}. \tag{2}$$

This is the same expression as (1).

Additionally, we adopt the convention $\sigma_{ij|l} = 0$ for all i, j with $v_i = v_j$.

Lane assignment

Assume now that class i wants to use lane l at rate $\lambda'_{i\to l}$ and that there are already other vessel classes $j < i$, all of higher priority, using that lane. Apart from congestion (considered immediately below), the rate $\lambda_{i\to l}$ at which class i actually gets to use lane l then depends on what share of its time it does not spend spent overtaking or yielding to vessels of higher priority:

$$\lambda_{i\to l} \quad = \quad \lambda'_{i\to l} \cdot \left(1 - \sum_{j<i} \sigma_{ij|l}\right). \tag{3}$$

Holding on to the assumption of constants speeds, the only congestion effect considered here is a space limitation on the amount of vessels on a lane. For this, consider a segment of length x on lane l. The expected number of class-i vessels on this segment is computed by noting that it takes a vessel x/v_i time units to traverse the segment, meaning that there are on average $\frac{x}{v_i}\lambda_{i\to l}$ vessels in the segment. With each vessel occupying d_i distance units, the share of space occupied by class-i vessels is

$$\varrho_{i\to l} \quad = \quad \frac{\lambda_{i\to l} d_i}{v_i}. \tag{4}$$

With $\sum_j \varrho_{j\to l}$ the share of space already occupied by class-j vessels, a share of $1-\sum_j \varrho_{j\to l}$ is available for class i, requiring

$$\lambda_{i\to l} \quad \leq \quad \frac{v_i}{d_i}\left(1-\sum_j \varrho_{j\to l}\right). \tag{5}$$

Including this in (3) yields the rate at which vessel class i is assigned to lane l:

$$\lambda_{i\to l} \quad = \quad \min\left\{\lambda'_{i\to l}\cdot\left(1-\sum_j \sigma_{ij|l}\right),\ \frac{v_i}{d_i}\left(1-\sum_j \varrho_{j\to l}\right)\right\}. \tag{6}$$

An incremental loading of all vessel classes onto the (one-directional part of a) fairway can now be given. The priority of a vessel class is given by its draught, in that vessels with largest draught have highest priority. The fairway is loaded sequentially with all vessel classes, from high to low priority. This ensures that when loading vessel class i onto the fairway, no vessel class $j<i$ has to react to class i. In addition, defining priority in terms of draught and loading the innermost lanes first allocates vessels with larger draught closer to the center of the fairway. The process looks as follows:

1. Label the vessel classes i such that class 1 has largest and class J has smallest draught.
2. Initialization. For all lanes $l=1,2,\ldots$:
   (a) $\varrho_l \leftarrow 0$.
   (b) for all vessel classes $i=1,\ldots,J$:
      i. $\lambda_{i\to l}\leftarrow 0$.
3. Incremental loading. For all vessel classes $i=1,\ldots,J$:
   (a) $\lambda'_{i\to 1}\leftarrow \lambda_i$.
   (b) $l=1$.
   (c) While $\lambda'_{i\to l}>0$, do:

$$\begin{aligned}
\lambda_{i\to l} &\leftarrow \min\left\{\lambda'_{i\to l}\cdot\left(1-\sum_{j=1}^{i-1}\sigma_{ij|l}\right),\ \frac{v_i}{d_i}\left(1-\varrho_l\right)\right\}\\
\varrho_l &\leftarrow \varrho_l+\frac{\lambda_{i\to l}d_i}{v_i}\\
\lambda'_{i\to 1+1} &\leftarrow \lambda'_{i\to l}-\lambda_{i\to l}\\
l &\leftarrow l+1.
\end{aligned}$$

Table 1: Passage usage statistics

	Passage 6		Passage 7	
	inbound	outbound	inbound	outbound
Compass directions	E, NE	W, SW	E, NE, SE	W, NW, SW
Total observations	661	651	290	316
Complete observations	355	316	67	74

Modeling of required fairway geometry

In overtaking situations, it is assumed that all lanes left of the rightmost (overtaking) vessel are currently used. If that was not the case then there would be an empty lane left of a used lane, contradicting the assumptions that vessels stay as far left as possible.

Given that lane l is used, its width W_l is modeled as a distributed quantity, where the probability of observing width w is

$$\Pr(W_l = w) \quad = \quad \sum_{i=1}^{J} \mathbf{1}(w_i = w) \frac{\varrho_{i \to l}}{\sum_{j=1}^{J} \varrho_{j \to l}}. \tag{7}$$

One may use different percentiles of this distribution to express more or less optimistic assumptions about the widths of used lanes and the resulting fairway geometry. The experiments reported further below consider an optimistic case where the 50ths width percentile is assumed for all lanes, and a pessimistic case where the 100th width percentile (i.e. the maximum width of all vessel classes using the lane) is used. The required depth of a lane is always set to the draught of the deepest vessel using that lane.

The percentile-based model aims to approximate that a vessel may be more inclined to overtake (on the right) if the vessel in front is located relatively far on the left (because it has relatively narrow vessels to its own left). This alone, however, may lead to infeasible fairway geometry estimates where certain vessels would not at all be able to traverse the fairway. A minimum fairway geometry is therefore imposed, which ensures for all width/draught pairs (w_i, g_i), $i = 1, \ldots, J$, that the width of the fairway part being at least g_i deep is at least w_i.

3 Case study

The two fairways outside of the port of Gothenburg shown in Figure 2 are considered. A data set covering all AIS-equipped vessel traffic in the study region during July 2023 is used [1,2]. All vessel movements crossing the two indicated passages either towards or away from the port are filtered out. Table 1 displays the usage statistics per passage, where a vessel movement is considered complete if it contains the following information: vessel type, vessel dimension (draught, width, length), speed over ground (subsequently just "speed").

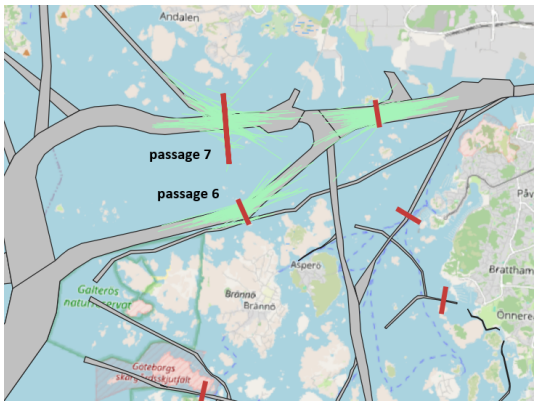


Figure 2: Gothenburg study area

Classification of vessels

Vessels are classified by draught and speed. Vessel segments are not used in the model and are therefore omitted from Tables~2 Table 3. Vessels moving towards (inbound) and away from (outbound) the port are classified separately.

All vessels within a class are treated as if they had the same draught (maximum value of draught interval) and speed (mean value of speed interval). The draught and interval sizes used in the classification could be chosen differently. Making the speed classes wider would mean that a wider range of vessels synchronizes towards the same speed; making the speed classes smaller would represent more speed variability and, hence, more aggressive maneuvering. The resolution of the draught discretization could be refined if correspondingly precise depth charts were available.

Scenario studies

The model is applied under different assumptions about the traffic volumes in the considered fairways. Base case traffic flow rates were obtained by dividing the number of vessels per (class, fairway, direction) with the observation period length of one month. Flow rates are uniformly scaled up to compensate for the exclusion of incomplete observations.

The effective width of a vessel is set to three times its physical width (meaning that the safety distance between two parallel moving vessels of the same class is two times their physical with). The effective length of a vessel is set to five times its length (meaning that a vessel keeps four time its length as a safety gap to the vessel in front). These size parameters could be chosen differently and, in particular, as functions of vessel inertia and speed.

Three scenarios are considered: The first scenario computes required fairway geometries based on actual fairway usages (base case). The second scenario as-

Table 2: Vessel classes in passage 6

Inbound

Draught [m]	Speed [km/h]	Market share [%]
12	27	0.56
12	15	0.28
10	15	0.56
10	27	1.69
10	21	2.54
8	27	14.93
8	33	17.18
8	15	2.25
8	21	10.42
8	9	0.85
6	33	17.18
6	39	4.79
6	27	5.35
6	21	6.20
6	15	8.73
6	9	1.69
4	9	0.56
4	21	1.69
4	15	2.54

Outbound

Draught [m]	Speed [km/h]	Market share [%]
12	21	0.32
10	27	1.27
10	21	4.43
8	27	22.15
8	33	9.18
8	21	9.81
8	15	2.22
8	9	0.32
6	33	21.84
6	39	0.63
6	27	7.28
6	21	5.70
6	15	8.86
6	9	0.32
4	21	1.27
4	15	4.43

Table 3: Vessel classes in passage 7

Inbound		
Draught [m]	Speed [km/h]	Market share [%]
16	9	1.49
14	9	4.48
10	15	1.49
10	33	1.49
10	21	1.49
8	15	1.49
8	33	5.97
8	27	5.97
8	21	1.49
6	33	2.99
6	9	1.49
6	21	2.99
4	15	2.99
4	21	58.21
4	27	1.49
2	15	4.48
Outbound		
Draught [m]	Speed [km/h]	Market share [%]
12	15	2.70
12	21	4.05
12	27	4.05
10	15	5.41
10	27	2.70
10	21	4.05
8	27	12.16
8	33	5.41
8	21	2.70
6	33	6.76
6	39	1.35
6	21	4.05
6	15	2.70
4	21	41.89

sumes all traffic is sent through a single fairway. The third scenario additionally imposes a speed limit of 10 km/h on the single used fairway. Figure 3 visualizes all results. Its left and right column considers Passage 6 and 7, respectively. Rows one, two, and three refer to one scenario each.

All subfigures display depth/draught [m] over lateral displacement [m], with the zero lateral coordinate indicating the interface between in- and outbound traffic. Lateral coordinates are from outbound (negative values) to inbound (positive), meaning that one looks towards the port.

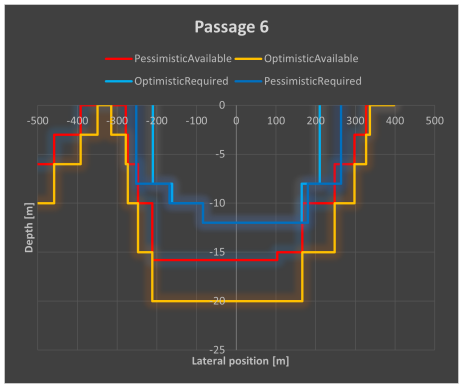
Each plot contains four lines. The red and orange lines represent a pessimistic and optimistic estimate of the available fairway depth, respectively; the red line corresponds to the DRVAL1 field in the depth chart; the orange line corresponds to the DRVAL2 field. The dark and light blue lines represent a pessimistic and optimistic of the required fairway depth, respectively: The dark blue line is obtained by using the 100th lane width percentile, the light blue line uses the 50th lane width percentile. The lines representing the fairway geometry (orange, red) are laterally shifted in the plots such that the required fairway profile (blue) is centered in the available fairway.

A situation where the dark blue line (pessimistic draught estimate) is above the red line (pessimistic depth estimate) may be interpreted as safe, while a situation where the light blue line (optimistic draught estimate) is below the orange line (optimistic depth estimate) may be labeled as unsafe. Borderline situations where neither one nor the other situation arises may indicate that the fairway operates at its safety limits. However, given their model-based construction, the labeling of as "safe" or "unsafe" need to be used with care. The labels may represent aspects of reality if the underlying model has been calibrated against that reality. More instructive than absolute safety levels is the model-predicted change of safety levels in reaction to scenario changes. Overall, given the at most intuitively calibrated model, the following scenario study is more a model illustration than a quantitative safety analysis.

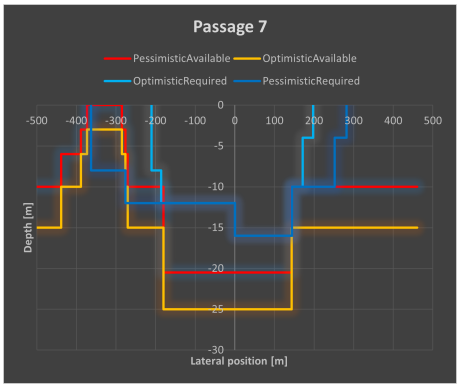
In the base case scenario, both passages appear well used. Figure 3(a) suggests that passage 6 is overall safe, while Figure 3(b) suggests that the outbound traffic in (the left part of) passage 7 is relatively more restricted by the fairway. The reason for this are the deep inbound vessels, which have to stay in the middle of the fairway, requiring the outbound traffic to push through a relatively narrow part of the passage. This suggests a relatively high risk of groundings and (due to the need to maneuver in more constrained space) collisions in the outbound part of passage 7.

Figures 3(c) and (d) illustrate a situation where one fairway is closed and all traffic is sent through the other fairway. One observes reduced safety levels in either fairway. The required fairway geometry grows less than proportionally with the amount of traffic because of the similar vessel classes operating in both fairways: Increasing the number of vessels traveling at the same speed does not induce lateral overtaking or yielding maneuvers among these vessels.

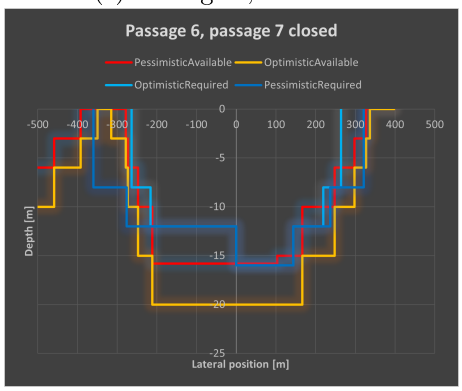
Figures 3(e) and (f) show what happens if all traffic is sent through a single fairway, combined with a speed limit of 10km/h imposed on all vessels. Both passages become "safe", with the sole exception being the insufficient depth of



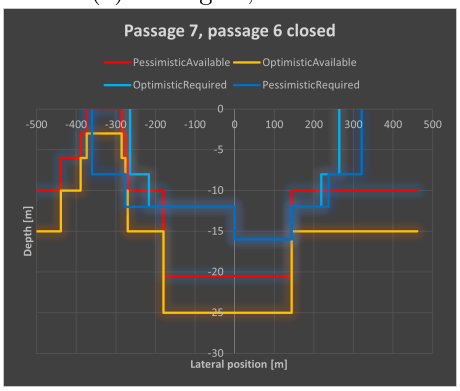
(a) Passage 6, base case



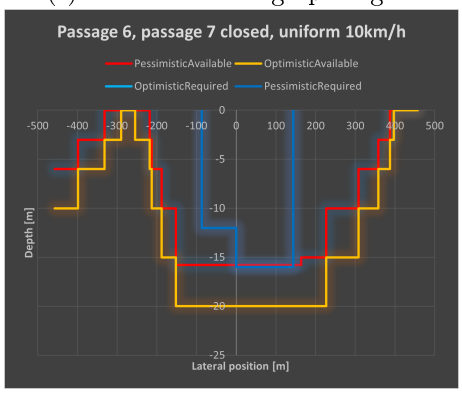
(b) Passage 7, base case



(c) All traffic through passage 6



(d) All traffic through passage 7



(e) Like (c), with 10 km/h speed limit



(f) Like (d), with 10 km/h speed limit

Figure 3: Case study results

passage 6 to accommodate the very deep vessels normally using passage 7. This last scenario illustrates that much of the lateral spread predicted by the model is the result of lateral maneuvers that can be reduced if vessels travel at more similar speeds.

4 Summary

The model may be of practical interest to analysts and planners assessing fairway capacity and evaluating the effects of changes in vessel traffic or fairway geometry. It can be used to study traffic-management measures that affect vessel ordering, routing, speed, or the allocation of fairway space.

The case study illustrates how observed vessel traffic can be translated into lateral space requirements and how these requirements can be compared with the available fairway geometry. In this way, the model provides a relatively simple means of investigating how changes in traffic demand, traffic composition, or traffic management affect the use of available fairway space.

The results should be interpreted in light of the simplifying assumptions of the model. In particular, the model represents lateral congestion through prioritized vessel classes and does not attempt to reproduce individual vessel movements or detailed interactions between vessels. Its intended use is therefore primarily for comparative and strategic analyses rather than for detailed simulation of vessel operations. The model can be used to compute overtaking rates as an additional indicator for assessing the plausibility of the predicted lateral space requirements.

The model determines the lateral space requirement at a given fairway section but does not determine the longitudinal extent over which this requirement applies. The fairway geometry and depth data used in an application must therefore be representative of the section being analysed.

References

[1] Danish Maritime Authority. (2025). Danish Maritime Authority Historical AIS data [Data set]. http://web.ais.dk/aisdata/. (Accessed October 2025).
[2] S&P Global Market Intelligence. (2024). Bespoke Data Catalogue: SHIPS [Data set]. Unpublished data (custom data extract commissioned by VTI, received June 2024).